DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL
FINAL EXAMINATION
Course Title: Data Structures
Course Code: CSE-1201
2nd Semester
Session: 2018-19

Time: 3 Hours **Marks:** 60

(Answer any FIVE questions)

1. a) Define data structure. Write down the reasons of studying data structure in Computer Science and Engineering. [2]

   b) Write down the algorithm for Insertion sort and analyze the best case and worst case running time for the algorithm. [3]

   c) Illustrate the operation of **HEAPSORT** on the Array, A= (5, 13, 2, 25, 7, 17, 20, 8, 4). [5]

   d) Write down the worst case and average case running time of Counting sort and Merge sort. [2]

2. a) What is recursion? Write a recursive algorithm to sum natural numbers (1, 2, 3 ..., N). [3]

   b) Define a Linked List. Write a function to insert an element in the middle of a Linked List. Let consider the following linked list with 12 as the head. Now write an algorithm to insert a new node (15) in the middle of node 18 and 7. [3]

   c) Sort the items using Quick Sort algorithm: 44, 33, 11, 55, 77, 44, 40, 90 and 60. [5]

   d) Give some examples of linear and non-linear data structure. [1]

3. a) Write down the number of hits does the following string matching algorithms encounter in the text ***T*= *abababacba*** when looking for the pattern ***P*= *ababac*** and also write down their procedure. [3]

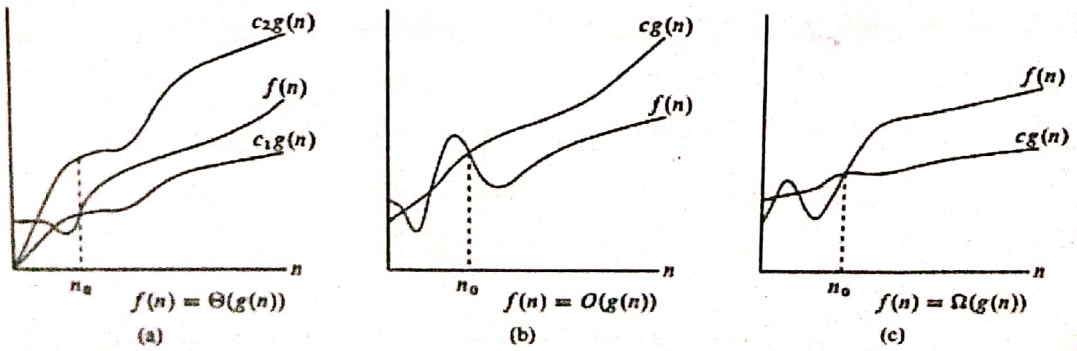
      i. Naive String Matching

   b) Consider two sequences: $X = (A, B, C, B, A, A, B)$ and $Y = (B, D, C, A, B, A)$. Find a maximum length common subsequence of X and Y. [**Hints:** Solve the problem using dynamic programming approach]. [5]

   c) How can you differentiate hash table from the direct addressing table? [2]

   d) Write down the recursive formula for "Tower of Hanoi" problem. [2]

4. a) What do you mean by growth of function? Briefly describe the following (a), (b) and (c) of the figure. [4]



   b) Define divide and conquer. How to find the result of 5^{57} % 7 using divide and conquer. [3]

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)

c) [5]

```
HOARE-PARTITION(A, p, r)
1   x = A[p]
2   i = p - 1
3   j = r + 1
4   while TRUE
5       repeat
6           j = j - 1
7       until A[j] ≤ x
8       repeat
9           i = i + 1
10      until A[i] ≥ x
11      if i < j
12          exchange A[i] with A[j]
13      else return j
```

Demonstrate the operation of HOARE-PARTITION on the array A = (13, 19, 9, 5, 12, 8, 7, 4, 11, 2, 6, 21), showing the values of the array and auxiliary values after each iteration of the for loop in lines 4-11.

5. a) "The average number of comparison required to find the location of an item is approximately equal to half the number of elements in an array"– justify the statement. ? [2]

b) Consider the following deque of characters where DEQUE is a circular array which allocates six memory cells: [4]

LEFT = 2, RIGHT = 4, DEQUE: ------, A, C, D, ------, ------.

Describe the deque while the following operations take place:

i) F is added to the right of the deque.

ii) Two letters on the right are deleted.

iii) K, L, M are added to the left of the deque.

iv) One letter on the left is deleted.

v) R is added to the left of the deque.

c) Consider a hash table of size 7 with hash function $h(k) = k \bmod 7$. Draw the table that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below: [6]

i) When collisions are handled by separate chaining.

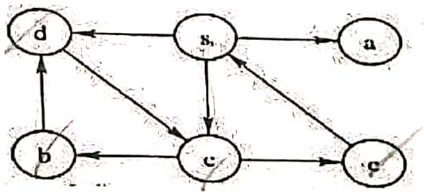
ii) When collisions are handled by linear probing.

iii) When collisions are handled by double hashing.

Using a second hash function $h'(k) = 5 - (k \bmod 5)$.

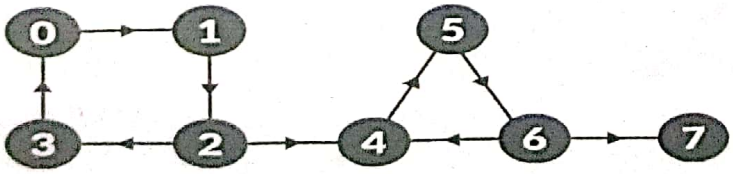
6. a) Is graph a data structure? Justify your answer. [2]

b) Draw the adjacency and incidence matrix for the above graph. Write down the advantages of adjacency matrix over adjacency list for graph traversal? [3]



c) Give the visited node order for breadth first search, starting with the source 's' from the graph depicted in question 6 (a). [3]

d) Find out the cycles from the graph using strongly connected components with appropiate diagram. [4]



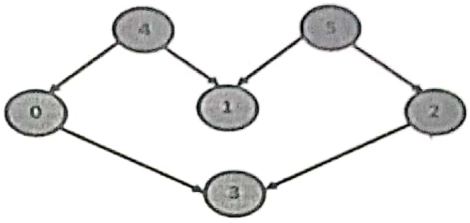
AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)

7. a) Consider the following adjacent list for directed graph. [5]

adj(y) = [x],
adj(x) = [z],
adj(z) = [y,w],
adj(w) = [x],
adj(s) = [z,w],
adj(v) = [s,w],
adj(t) = [u,v],
adj(u) = [v].

i. Construct a graph from the above information.
ii. Identify the tree edge, back edge, cross edge and forward edge for the graph

b) Topological sort algorithm for Directed Acyclic Graph (DAG) is a linear arrangement of vertices such that for every directed edge x-y, vertex x comes before y in the arrangement. Topological sorting is only applicable on graphs that are DAG. Now your task is to prove that from the above [5]

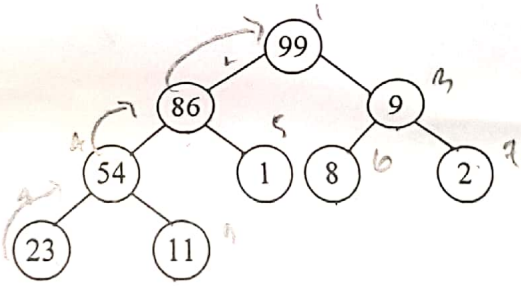


graph that, "There can exist more than one topological sorting for a given graph."

c) "Is bubble sort is divide and conquer type" Explain your answer if you say 'yes' or give a counter example if you say 'no'? [2]

8. a) What is recursion? Write a recursive algorithm to sum natural numbers (1, 2, 3 ..., N). [3]

b) Consider the following heap: [5]



For each of the following questions use the original heap. Do not use the heap obtained after each answer.

i) Represent the heap as an array indexed by 1 through 9.
ii) Delete the element 99 from the heap shown in the above figure.
iii) Delete the element 23 from the heap shown in the above figure.
iv) Delete the element 8 from the heap shown in the above figure.
v) Is it max heap or min heap? Sorting the elements in descending order by heap sort.

c) What kind of problem can be solved using dynamic programming algorithm. [2]

d) How to convert a queue into the stack with the procedure and examples? [2]

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)